Tarea 2 IPD432

*Implementación de un co-procesador en FPGA

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***Abstract*—**

I. Introduction

El siguiente documento contiene la descrpición de la implementación de un co-procesador en la tarjeta Nexys A7-100T, el cual es capaz de ejecutar distintas operaciones artitmeticas sobre vectores de 1024 valores de 11 bits. Para esto, el co-procesador o *device* se comunica con un *host* a través de una interfaz UART configurada a 115200 baudios. El *device* es capaz de leer los datos almacenados en bloques de memoria de la FPGA, calcular la suma, promedio, distancia de manhattan, euclidiana y producto punto de los vectores, y finalmente enviar los resultados al *Host* a través de la misma interfaz UART. La implementación del la tarea se puede dividir en dos aspectos, la **Interfaz de usuario** y la **Lógica de procesamiento**.

II. Interfaz

La interfaz de usuario se encarga de facilitar la interacción del *host* con el *device* y esconder la complejidad del sistema. Para esto, se escribe un script de Matlab con funciones que permiten enviar los datos de entrada y recibir los resultados de las operaciones. Las funciones principales implementadas son:

A. write2dev

La función write2dev(vector, bloque, puerto) le permite al usuario enviar y guardar un vector de 1024 palabras de 11 bits en el bloque de memoria que se eliga. Puesto que la UART solo puede enviar paquetes de 8 bits (1 byte), la función envía dos bytes por cada dato del vector, los cuáles serán re-armados por la lógica implementada por la FPGA.

B. command2dev

La función command2dev(comando , args) recibe como primer argumento el nombre de los comandos implementados, y como segundo argumendo un arreglo de datos necesarios para poder ejecutar el comando de manera exitosa, el detalle de documentación para la función se encuentra en el III-B

III. Lógica de procesamiento

La lógica del *device* se construye a partir del taller de IPs visto en clases, el circuito planteado en esa instancia permitía recibir y guardar palabras de un byte en memoria y luego enviarlas de vuelta al *host* cuando sea solicitado. Para cumplir con los requerimientos del enunciado, el *device* debe:

- Ser capaz de guardar vectores de 1024 palabras de 11 bits en dos bloques de memoria.
- Reconocer los comandos recibidos por el *host* y enviar las señales necesarias a la unidad de cálculo.
- Operar sobre los vectores guardados, leyendo desde los bloques de memoria y operando segun las instrucciones del item previo.
- Enviar los datos requeridos de vuelta al *host* y mostrarlos a traves de los periféricos de la placa en los casos necesarios.

Para lograr lo anterior, el enunciado propone el siguiente diagrama de alto nivel. Para cumplir con lo anterior, se

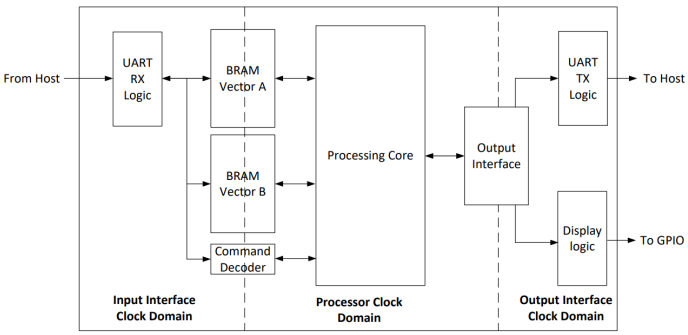


Fig. 1. Diagrama propuesto

diseñaron los bloques *Command Decoder*, *Processing Core* y *Display logic*.

A. Command Decoder

El módulo *Command Decoder* monitorea los bytes recibidos desde la UART a través de `rx_data` cuando se activa el pulso `rx_ready`. Mediante la señal interna `cmd_valid`, verifica si el byte recibido corresponde a un comando de lectura u operación (valores del 2 al 8) mientras las memorias no están siendo escritas (`rx_busy = 1'b0`). Si el comando

corresponde a una cabecera de escritura en memoria (0 o 1) o si *RX_Logic* se encuentra cargando un vector (`rx_busy = 1'b1`), entonces `cmd_valid = 1'b0` para evitar que los datos del vector sean interpretados como instrucciones. Cuando `cmd_valid = 1'b1`, el módulo registra el código de la instrucción en `op_mode[2:0]` y genera un pulso de un ciclo de reloj en la señal `start_op` para indicar al *Processing Core* que inicie el procesamiento. Adicionalmente, si el comando corresponde a una operación de reducción escalar, activa la bandera respectiva (`is_man_dist`, `is_euc_dist` o `is_dot_prod`). Estas banderas permiten habilitar el acumulador en el *Processing Core*, seleccionar la cantidad de bytes a enviar en *TX_logic* (2, 3 o 4 bytes) y encender los displays de 7 segmentos en *Display_logic*.

B. Abbreviations and Acronyms

Define abbreviations and acronyms the first time they are used in the text, even after they have been defined in the abstract. Abbreviations such as IEEE, SI, MKS, CGS, ac, dc, and rms do not have to be defined. Do not use abbreviations in the title or heads unless they are unavoidable.

C. Units

- Use either SI (MKS) or CGS as primary units. (SI units are encouraged.) English units may be used as secondary units (in parentheses). An exception would be the use of English units as identifiers in trade, such as "3.5-inch disk drive".
- Avoid combining SI and CGS units, such as current in amperes and magnetic field in oersteds. This often leads to confusion because equations do not balance dimensionally. If you must use mixed units, clearly state the units for each quantity that you use in an equation.
- Do not mix complete spellings and abbreviations of units: "Wb/m^2" or "webers per square meter", not "webers/m^2". Spell out units when they appear in text: ". . . a few henries", not ". . . a few H".
- Use a zero before decimal points: "0.25", not ".25". Use "cm^3", not "cc".)

D. Equations

Number equations consecutively. To make your equations more compact, you may use the solidus (/), the exp function, or appropriate exponents. Italicize Roman symbols for quantities and variables, but not Greek symbols. Use a long dash rather than a hyphen for a minus sign. Punctuate equations with commas or periods when they are part of a sentence, as in:

$$a + b = \gamma \tag{1}$$

Be sure that the symbols in your equation have been defined before or immediately following the equation. Use "(1)", not "Eq. (1)" or "equation (1)", except at the beginning of a sentence: "Equation (1) is . . ."

E. LaTeX-Specific Advice

Please use "soft" (e.g., `\eqref{Eq}`) cross references instead of "hard" references (e.g., `(1)`). That will make it possible to combine sections, add equations, or change the order of figures or citations without having to go through the file line by line.

Please don't use the `{eqnarray}` equation environment. Use `{align}` or `{IEEEeqnarray}` instead. The `{eqnarray}` environment leaves unsightly spaces around relation symbols.

Please note that the `{subequations}` environment in LaTeX will increment the main equation counter even when there are no equation numbers displayed. If you forget that, you might write an article in which the equation numbers skip from (17) to (20), causing the copy editors to wonder if you've discovered a new method of counting.

BibTeX does not work by magic. It doesn't get the bibliographic data from thin air but from .bib files. If you use BibTeX to produce a bibliography you must send the .bib files.

LaTeX can't read your mind. If you assign the same label to a subsubsection and a table, you might find that Table I has been cross referenced as Table IV-B3.

LaTeX does not have precognitive abilities. If you put a `\label` command before the command that updates the counter it's supposed to be using, the label will pick up the last counter to be cross referenced instead. In particular, a `\label` command should not go before the caption of a figure or a table.

Do not use `\nonumber` inside the `{array}` environment. It will not stop equation numbers inside `{array}` (there won't be any anyway) and it might stop a wanted equation number in the surrounding equation.

F. Some Common Mistakes

- The word "data" is plural, not singular.
- The subscript for the permeability of vacuum μ_0, and other common scientific constants, is zero with subscript formatting, not a lowercase letter "o".
- In American English, commas, semicolons, periods, question and exclamation marks are located within quotation marks only when a complete thought or name is cited, such as a title or full quotation. When quotation marks are used, instead of a bold or italic typeface, to highlight a word or phrase, punctuation should appear outside of the quotation marks. A parenthetical phrase or statement at the end of a sentence is punctuated outside of the closing parenthesis (like this). (A parenthetical sentence is punctuated within the parentheses.)
- A graph within a graph is an "inset", not an "insert". The word alternatively is preferred to the word "alternately" (unless you really mean something that alternates).
- Do not use the word "essentially" to mean "approximately" or "effectively".
- In your paper title, if the words "that uses" can accurately replace the word "using", capitalize the "u"; if not, keep using lower-cased.

- Be aware of the different meanings of the homophones "affect" and "effect", "complement" and "compliment", "discreet" and "discrete", "principal" and "principle".
- Do not confuse "imply" and "infer".
- The prefix "non" is not a word; it should be joined to the word it modifies, usually without a hyphen.
- There is no period after the "et" in the Latin abbreviation "et al.".
- The abbreviation "i.e." means "that is", and the abbreviation "e.g." means "for example".

An excellent style manual for science writers is [7].

G. Authors and Affiliations

The class file is designed for, but not limited to, six authors. A minimum of one author is required for all conference articles. Author names should be listed starting from left to right and then moving down to the next line. This is the author sequence that will be used in future citations and by indexing services. Names should not be listed in columns nor group by affiliation. Please keep your affiliations as succinct as possible (for example, do not differentiate among departments of the same organization).

H. Identify the Headings

Headings, or heads, are organizational devices that guide the reader through your paper. There are two types: component heads and text heads.

Component heads identify the different components of your paper and are not topically subordinate to each other. Examples include Acknowledgments and References and, for these, the correct style to use is "Heading 5". Use "figure caption" for your Figure captions, and "table head" for your table title. Run-in heads, such as "Abstract", will require you to apply a style (in this case, italic) in addition to the style provided by the drop down menu to differentiate the head from the text.

Text heads organize the topics on a relational, hierarchical basis. For example, the paper title is the primary text head because all subsequent material relates and elaborates on this one topic. If there are two or more sub-topics, the next level head (uppercase Roman numerals) should be used and, conversely, if there are not at least two sub-topics, then no subheads should be introduced.

I. Figures and Tables

a) Positioning Figures and Tables: Place figures and tables at the top and bottom of columns. Avoid placing them in the middle of columns. Large figures and tables may span across both columns. Figure captions should be below the figures; table heads should appear above the tables. Insert figures and tables after they are cited in the text. Use the abbreviation "Fig. 1", even at the beginning of a sentence.

Figure Labels: Use 8 point Times New Roman for Figure labels. Use words rather than symbols or abbreviations when writing Figure axis labels to avoid confusing the reader. As an example, write the quantity "Magnetization", or "Magnetization, M", not just "M". If including units in the label, present them within parentheses. Do not label axes only with units. In the example, write "Magnetization (A/m)" or "Magnetization {A[m(1)]}", not just "A/m". Do not label axes with a ratio of quantities and units. For example, write "Temperature (K)", not "Temperature/K".

TABLE I
TABLE TYPE STYLES

Table Head	**Table Column Head**		
	Table column subhead	***Subhead***	***Subhead***
copy	More table copy[a]		

[a]Sample of a Table footnote.

ACKNOWLEDGMENT

The preferred spelling of the word "acknowledgment" in America is without an "e" after the "g". Avoid the stilted expression "one of us (R. B. G.) thanks ...". Instead, try "R. B. G. thanks...". Put sponsor acknowledgments in the unnumbered footnote on the first page.

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Please number citations consecutively within brackets [1]. The sentence punctuation follows the bracket [2]. Refer simply to the reference number, as in [3]—do not use "Ref. [3]" or "reference [3]" except at the beginning of a sentence: "Reference [3] was the first ..."

Number footnotes separately in superscripts. Place the actual footnote at the bottom of the column in which it was cited. Do not put footnotes in the abstract or reference list. Use letters for table footnotes.

Unless there are six authors or more give all authors' names; do not use "et al.". Papers that have not been published, even if they have been submitted for publication, should be cited as "unpublished" [4]. Papers that have been accepted for publication should be cited as "in press" [5]. Capitalize only the first word in a paper title, except for proper nouns and element symbols.

For papers published in translation journals, please give the English citation first, followed by the original foreign-language citation [6].

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